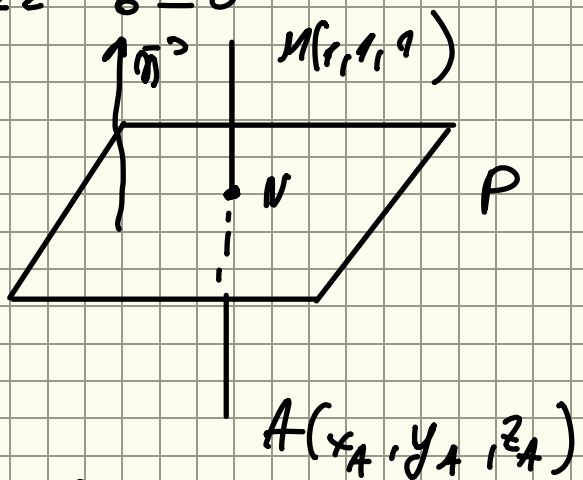
Seminar 4

L7.) Determine the point symmetric to $M(1,1,1)$ with respect to plane $P: x+y-2z-6=0$



$\vec{n} = \vec{i} + \vec{j} - 2\vec{k}$

$N = AM \cap P \Rightarrow A = 2N - M$

$\left.\begin{matrix} \vec{n} \perp P \\ AM \perp P \end{matrix}\right| \Rightarrow \vec{n} \parallel AM$

$\Rightarrow \begin{cases} x = 1+t \\ y = 1+t \\ z = 1-2t \end{cases}$, $N \in AM \Rightarrow \exists\, t \in \mathbb{R}$ s.t. $N(1+t, 1+t, 1-2t)$, $N \in P$ $\Rightarrow$

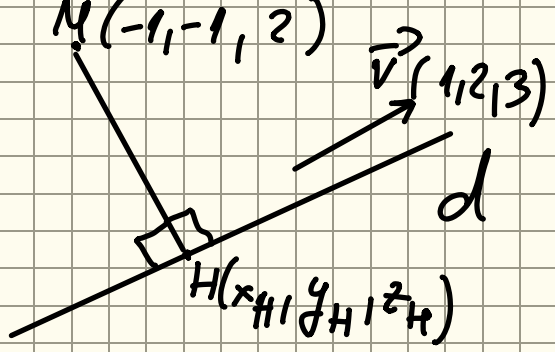
$1+t+1+t-2+4t-6=0$

$6t = 6 \Rightarrow t = 1$

$\Rightarrow N(2,2,-1)$

$\Rightarrow A(3,3,-3)$

L8.) Determine the orthogonal projection of the point $M(-1,-1,2)$ onto $d: \begin{cases} x = 2+t \\ y = -1+2t \\ z = 1+3t \end{cases}$, $t \in \mathbb{R}$



$H = d \cap d(M, d)$

$H \in d \Rightarrow \exists\, t \in \mathbb{R}$ s.t. $\begin{cases} x_H = 2+t \\ y_H = -1+2t \\ z_H = 1+3t \end{cases}$

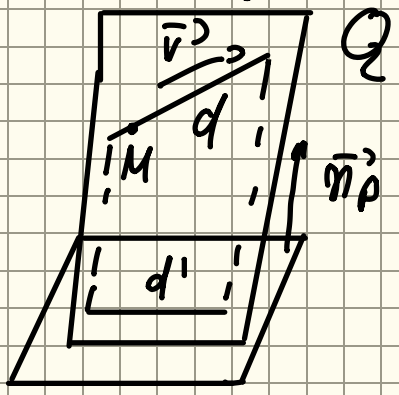
$\vec{v} = \vec{i} + 2\vec{j} + 3\vec{k}$

$\overrightarrow{MH} = \overrightarrow{OH} - \overrightarrow{OM} = (3+t, 2t, -1+3t)$

$\vec{v} \perp \overrightarrow{MH} \Rightarrow \vec{v} \cdot \overrightarrow{MH} = 0 \Rightarrow 3+t+4t-3+9t = 0$

$\Leftrightarrow 14t = 0 \Rightarrow t = 0 \Rightarrow H(2, -1, 1)$

Lo) Write the equation of the orthogonal projection of $d: \frac{x-3}{2} = \frac{y+1}{-2} = z-3 = t$ onto the plane $P: x - 3y + z + 9 = 0$



$\vec{n}_P = \vec{i} - 3\vec{j} + \vec{k}$

$\left.\begin{matrix} \vec{n}_P \perp P \\ Q \perp P \end{matrix}\right| \Rightarrow \vec{n}_P \parallel Q$

Q - plane and $d \in Q$

$Q \perp P$

$\left.\begin{matrix} \vec{v} \parallel d \\ d \subset Q \end{matrix}\right| \Rightarrow \vec{v} \parallel Q$

$\vec{v} = (2, -2, 1)$

$M(3, -1, 3) \in d \subset Q \Rightarrow$

$M \in Q$

$$\Rightarrow \begin{vmatrix} x-3 & y+1 & z-3 \\ 1 & -3 & 1 \\ 2 & -2 & 1 \end{vmatrix} = 3 - x + y + 1 + 4z - 12 = 0$$

$\Rightarrow Q: x - y - 4z + 8 = 0$

$$\Rightarrow \left.\begin{matrix} d \subset Q \\ d \subset P \end{matrix}\right| \Rightarrow \begin{cases} x - y - 4z = -8 \\ x - 3y + z = -9 \end{cases}$$

$$\bar{A} = \left[\begin{array}{ccc|c} 1 & -1 & -4 & -8 \\ 1 & -3 & 1 & -9 \end{array}\right] \overset{L_2 - L_1}{\sim} \left[\begin{array}{ccc|c} 1 & -1 & -4 & -8 \\ 0 & -2 & 5 & -1 \end{array}\right]$$

$$\Rightarrow z = \alpha \in \mathbb{R} \Rightarrow \begin{cases} x - y = 4\alpha - 8 \\ -2y = -1 - 5\alpha \end{cases}$$

$$\Rightarrow y = \frac{1+5\alpha}{2} \Rightarrow x = \frac{13\alpha - 15}{2}$$

$$\alpha = 1 \Rightarrow y = 3 \Rightarrow x = -1 \quad \Big| \Rightarrow A(-1, 3, 1)$$

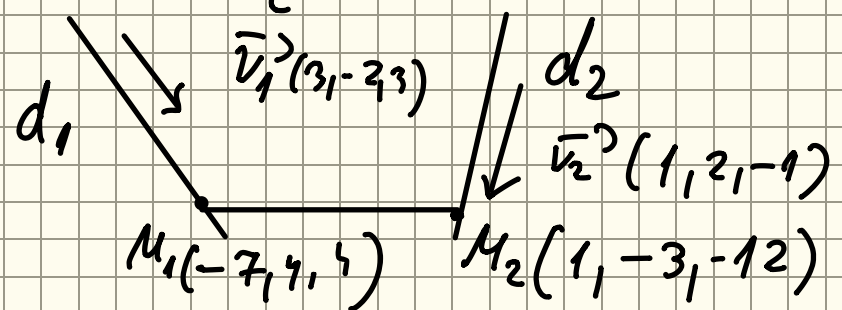
$$\alpha = -1 \Rightarrow y = -2 \Rightarrow x = -14 \quad \Big| \; B(-14, -2, -1)$$

$$\vec{AB} = (-13, 5, 2) \quad \Big| \Rightarrow \begin{cases} x = -14 - 13t \\ y = -2 + 5t \\ z = -1 + 2t \end{cases}$$

$B \in d$

L10) For each of the following pairs, prove that the lines are skew, compute the distance between them and write the equations of their common perpendicular.

a) $d_1: \begin{cases} x = 3t - 7 \\ y = -2t + 4 \\ z = 3t + 4 \end{cases}$, $d_2: \frac{x-1}{1} = \frac{y+3}{2} = \frac{z+12}{-1}$



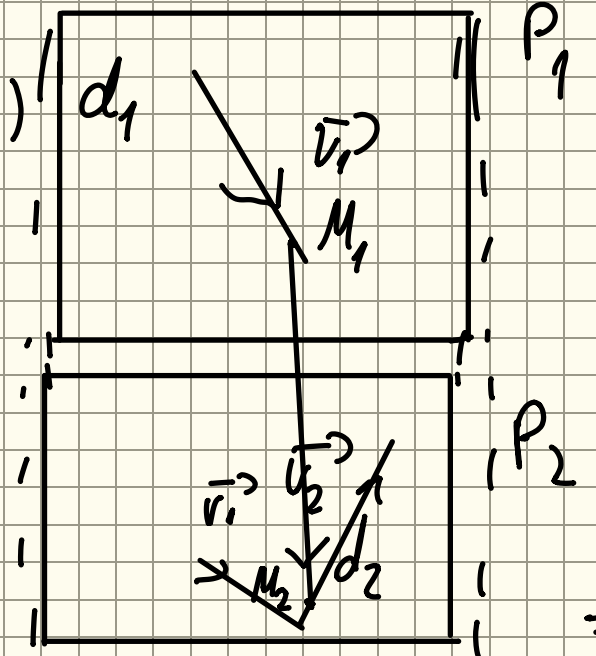
Suppose d_1, d_2 are coplanar

$\vec{v_1} \parallel d_1, \vec{v_2} \parallel d_2$
$M_1 \in d_1, M_2 \in d_2$ $\Rightarrow \overrightarrow{M_1M_2}, \vec{v_1}, \vec{v_2}$ are coplanar

$$\Rightarrow \begin{vmatrix} 3 & -2 & 3 \\ 1 & 2 & -1 \\ 8 & -7 & -16 \end{vmatrix} = 3\begin{vmatrix} 2 & -1 \\ -7 & -16 \end{vmatrix} + 2\begin{vmatrix} 1 & -1 \\ 8 & -16 \end{vmatrix} + 3\begin{vmatrix} 1 & 2 \\ 8 & -7 \end{vmatrix}$$

$= -202 \neq 0 \Rightarrow d_1, d_2$ cannot be coplanar

b)



$V_p = A_{base} \cdot d(d_1, d_2)$

$\Rightarrow d(d_1, d_2) = \frac{V_p}{A_{base}}$

$V_p = (\overrightarrow{M_1M_2}, \vec{v_1}, \vec{v_2})$

$A_{base} = |\vec{v_1} \times \vec{v_2}| = \sqrt{116}$

$\Rightarrow d(d_1, d_2) = \frac{|-202|}{\sqrt{116}}$

c) Let $A \in d, B \in d$ s.t. $AB \perp d_1, d_2$

$\Rightarrow A = (3t-7, -2t+4, 3t+4)$

$B = (1+s, 20-3, -s-12)$

$AB \perp d_1, \vec{v_1} \parallel d_1 \Rightarrow \overrightarrow{AB} \perp \vec{v_1} \Leftrightarrow \overrightarrow{AB} \cdot \vec{v_1} = 0$

$AB \perp d_2, \vec{v_2} \parallel d_2 \Rightarrow \overrightarrow{AB} \perp \vec{v_2} \Leftrightarrow \overrightarrow{AB} \cdot \vec{v_2} = 0$

$\overrightarrow{AB} = (8+s-3t, 20+2t-7, -s-3t-16)$

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